# Aadit Kannan

## Project Portfolio

This portfolio includes my hardware, electrical, and software projects, along with some of my physics and matsci work on semiconductor thin-films.

Some of my projects are included here. For the full list, go to aaditkannan.com/projects.


CONTACT

WEBSITE
aaditkannan.com

PROJECTS
aaditkannan.com/projects

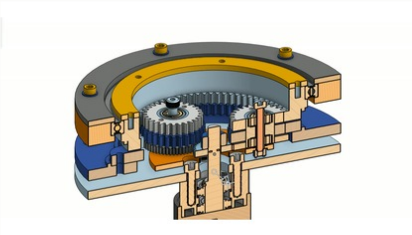
EMAIL
aaditkannan@berkeley.edu

GITHUB
github.com/aaditkannan

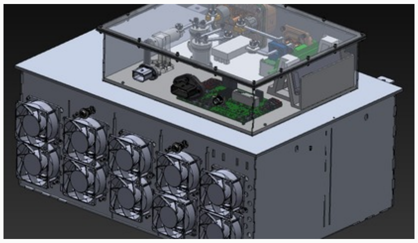
LINKEDIN
linkedin.com/in/aaditkannan


PORTFOLIO CONTENTS

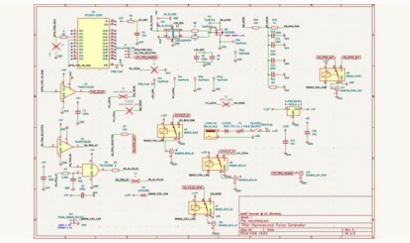




Actuator section render



Accumulator attic CAD

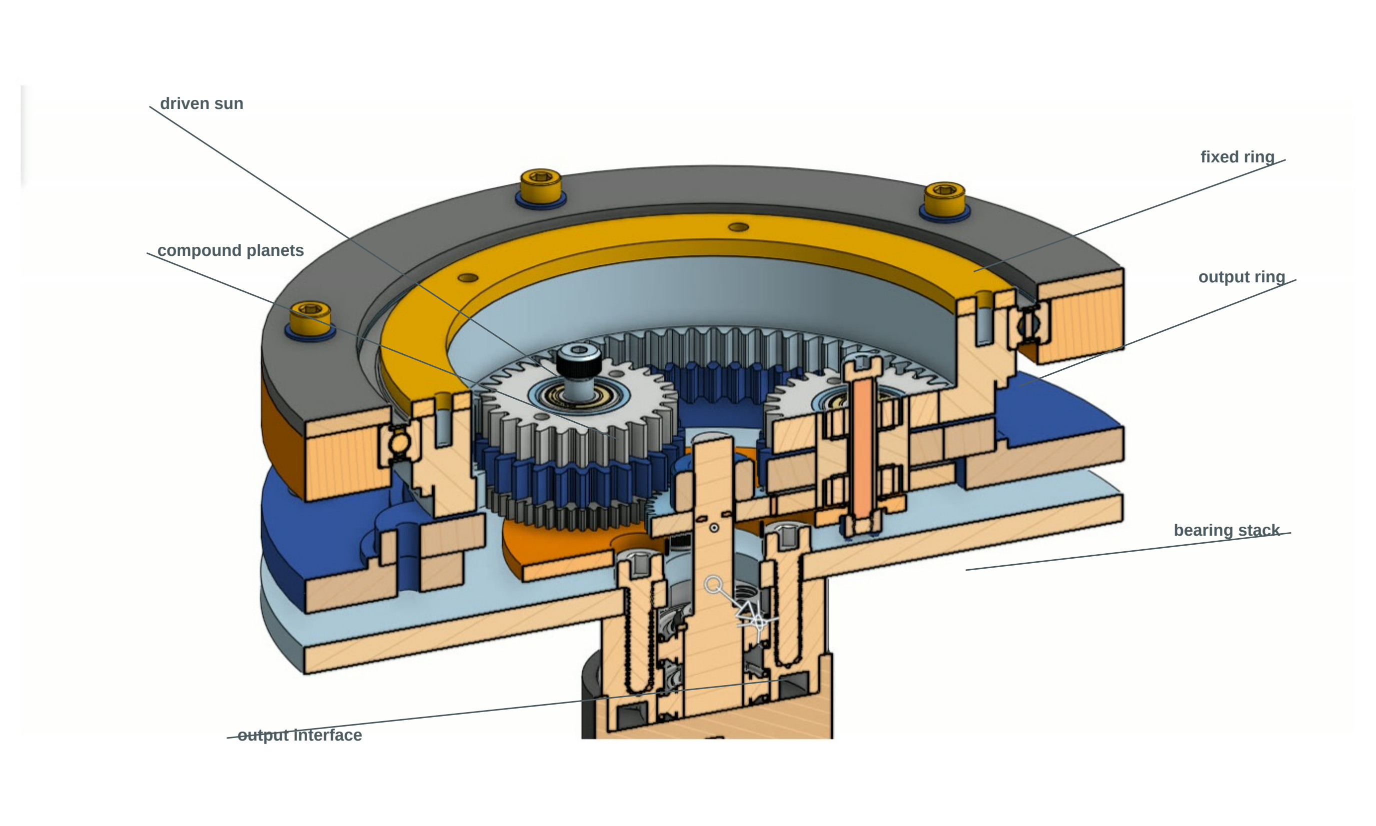


Pulse-generator schematic

# Wolfrom Robotic Actuator

Jun 2026 - Present / Robotics Hardware

Compact compound planetary actuator for humanoid-scale joints, focused on high reduction, backdrivability, reflected inertia, and mechanical packaging.




CAD section: driven sun, compound planets, fixed ring, output ring, bearing stack, and output interface.

### DESIGNED

- Full actuator stackup: sun, compound planet pairs, fixed ring, output ring, carrier, bearings, shafts, and output hub.
- One compact axial package instead of stacking multiple conventional planetary stages.
- Split output ring, bolt access, dowel alignment, bearing support, and assembly timing strategy.
- Printed prototype geometry first, with a metal revision path around CNC aluminum and wire-EDM ring gears.

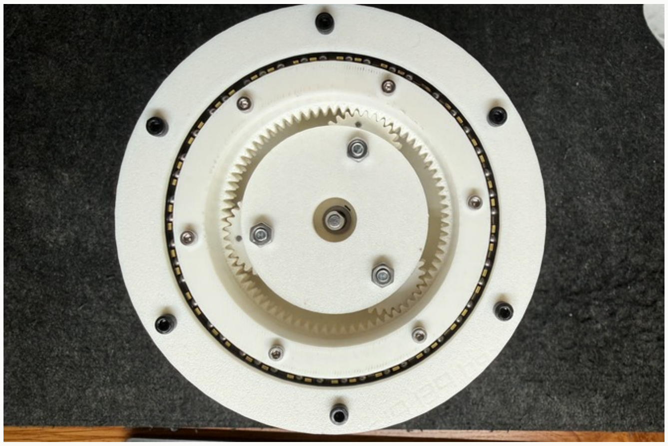
### TECHNICAL NOTES

- Current tooth counts produce 555/11, or about 50.45:1 reduction; a 125:1 variant is in the same design family.
- Ratio split moves more reduction into the sun/carrier stage and less into the lossy ring differential.
- Main loss paths are ring-mesh friction, planet-bearing drag, load sharing, and rough printed tooth surfaces.
- Backdrivability work focuses on reducing friction rather than pretending reflected rotor inertia disappears.

# Wolfrom Actuator - Prototype + Next Revision

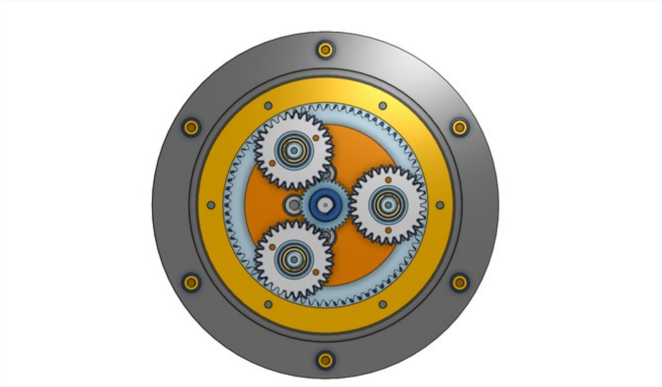
Jun 2026 - Present / Robotics Hardware

Printed prototypes are being used for gear mesh, packaging, stack height, and assembly checks before committing to the metal build.

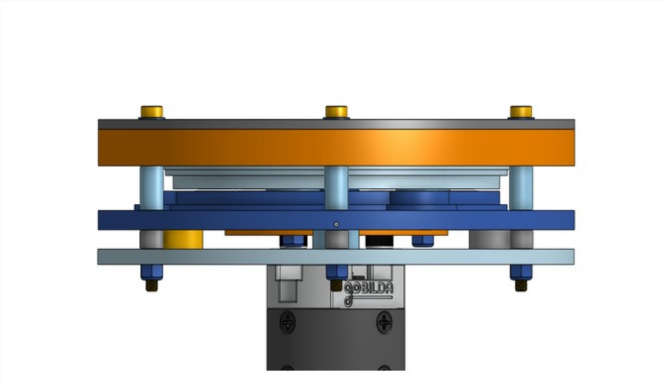


Printed prototype used for gear mesh, bearing placement, and stack-height checks.

Bench assembly and fit-checking before re-timing the actuator by hand.



Top view: three compound planets, output ring, fastener pattern, and center sun.



Side view: plate stack, motor interface, standoffs, and axial package.

### CURRENT PROTOTYPE

- Built and fit-checked printed actuator hardware for meshing, assembly access, and stack alignment.
- Working toward split-gear preloading for near-zero backlash; no final backlash claim yet.
- Evaluating printed gears as geometry prototypes, not as final load-bearing actuator hardware.

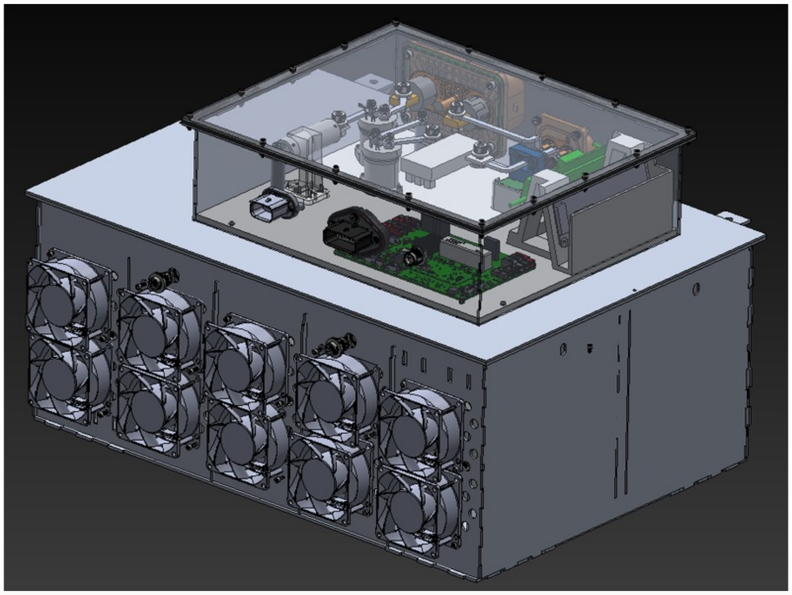
### NEXT REVISION

- KISSsoft refinement for macro-geometry, profile shifts, and tip relief on the loaded ring meshes.
- Metal version planned around CNC aluminum structure, wire-EDM hardened-steel rings, hardened pins, and needle bearings.
- Dyno characterization planned for breakaway torque, backdrive feel, efficiency, and load behavior after the metal revision.

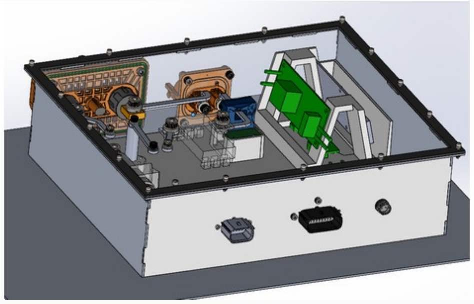
# Formula Electric 588V Accumulator

Sep 2025 - Present / High-Voltage Packaging

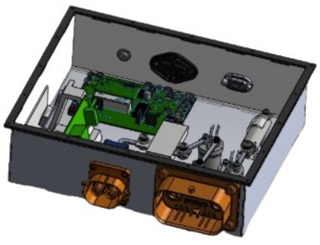
Designed and manufactured accumulator attic hardware for Berkeley Formula Electric: HV electronics packaging, enclosure design, busbar routing, waterproofing, insulation, cooling, and inspection access.



588V accumulator CAD showing the HV electronics attic, fan wall, service access, and enclosure packaging.



Electronics attic layout: board, connector, busbar, and mechanical interface packaging.



Internal attic CAD view with board/connector access and enclosure clearances.

### BUILT

- Packaged the HV electronics attic, busbars, boards, fans, fasteners, and service interfaces inside the accumulator envelope.
- Reduced attic mass by 18% while preserving insulation clearances, cooling paths, inspection access, and fastener access.
- Designed around a 588V pack, waterproofing needs, high-dielectric insulation, and SES/ESF review material.

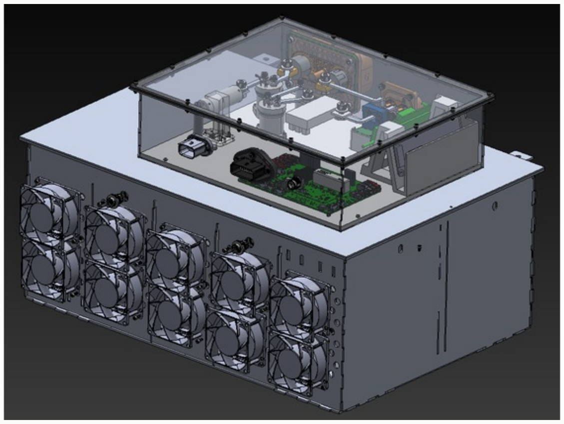
### CONSTRAINTS

- Busbar routing for 80A peak discharge without turning the attic into an unserviceable wiring box.
- FSAE structural load cases, high-voltage isolation, rain-test sealing, fan airflow, and board access all fight for the same volume.
- Transparent/inspectable surfaces and removable interfaces mattered because serviceability is part of safety.

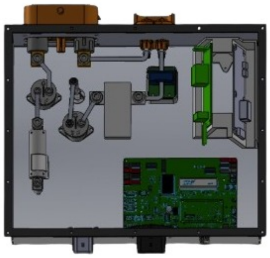
# Accumulator Hardware - Materials + Validation

Sep 2025 - Present / High-Voltage Packaging

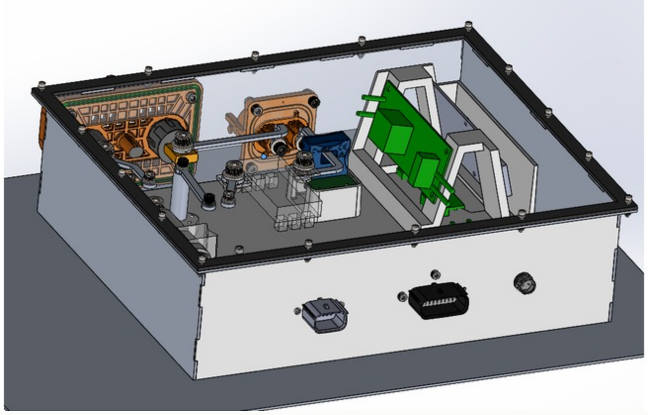
The accumulator attic is a packaging problem, a manufacturing problem, and a safety-review problem at the same time.



Overall accumulator CAD context: attic, enclosure, service access, and internal packaging.



Top-down attic CAD view used to check electronics layout and access.



Detailed accumulator subassembly view for board, connector, and internal support packaging.

### MANUFACTURING PLAN

- Laser-cut aluminum panels, TIG welded structure, waterjet-cut polycarbonate floor, and waterjet-cut neoprene gasket.
- Polycarbonate floor provides electrical isolation and visual inspection without disassembling the accumulator.
- Nomex 410 insulation selected for dielectric strength, high temperature rating, and flame resistance near HV components.

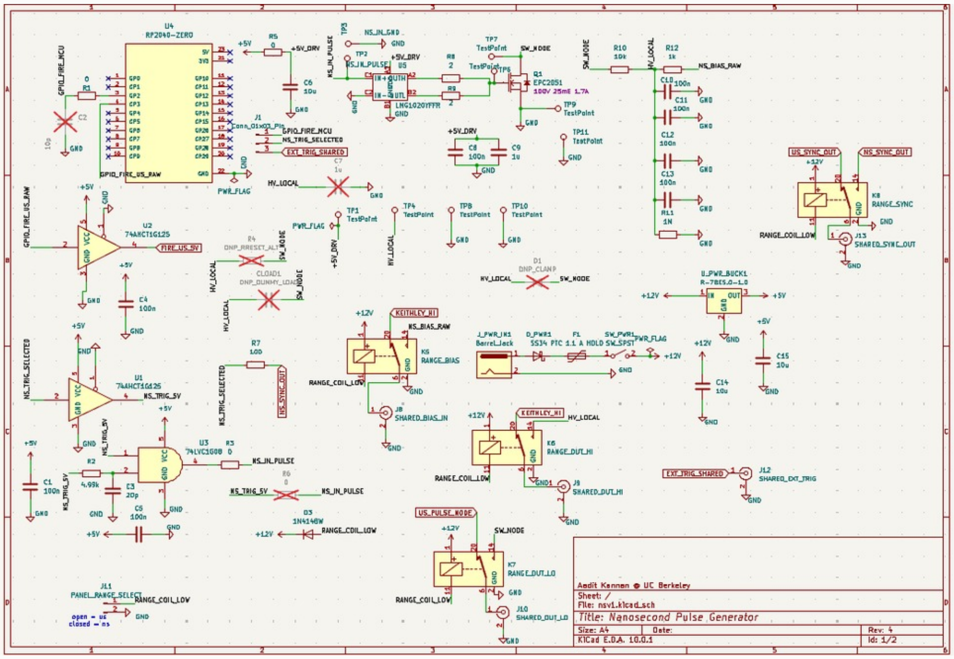
### CHECKS

- FEA against 40g lateral, 40g longitudinal, and 20g vertical load cases with yield margin.
- Dielectric margin, gasket compression, water ingress, service access, fastener access, and cooling airflow reviewed together.
- 1500V AC dielectric test and IP65-style water ingress testing are the target validation gates before competition use.

# Nanosecond Pulse Generator

May 2026 - Present / Research Hardware / PCB

Nanosecond and microsecond pulse-generator board for $BiFeO_3$ spin-transport and ferroelectric characterization with GaN switching, source-measure biasing, grounded coax/BNC I/O, and RP2040/Python triggering.




Current nanosecond pulse-generator schematic. Shown as the main artifact because it is the current board revision.

#### BENCH INTERFACE

Python/RP2040 trigger -> driver and switching stage -> coax/BNC output -> device or probe station -> oscilloscope and source-measure capture.

#### DESIGNED

- Dual-range pulse architecture around current BFO testing near 30V, with future sweeps toward smaller voltage ranges.
- GaN switching path, trigger/sync routing, grounded lab I/O, and source-measure bias interface.
- Board organization around probe-station use, scope validation, dummy-load bring-up, and controlled pulse visibility.

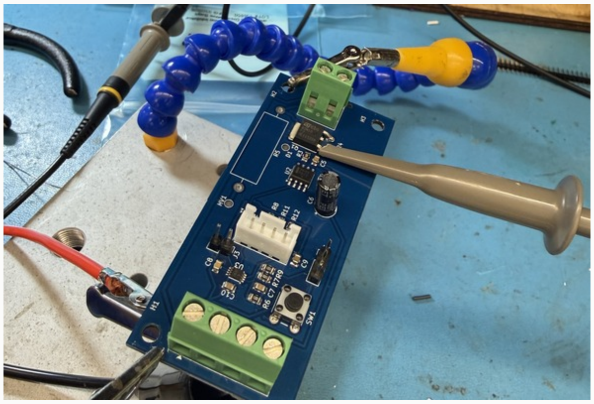
#### CASE-SPECIFIC NOTES

- The useful quantity is the voltage that actually lands across the BFO device, not just a nominal connector setting.
- Current response mixes polarization switching, leakage, dielectric capacitance, and fixture parasitics, so timing and grounding matter.
- No final rise-time, jitter, device result, or amplitude-stability claim yet; bring-up is still in progress.

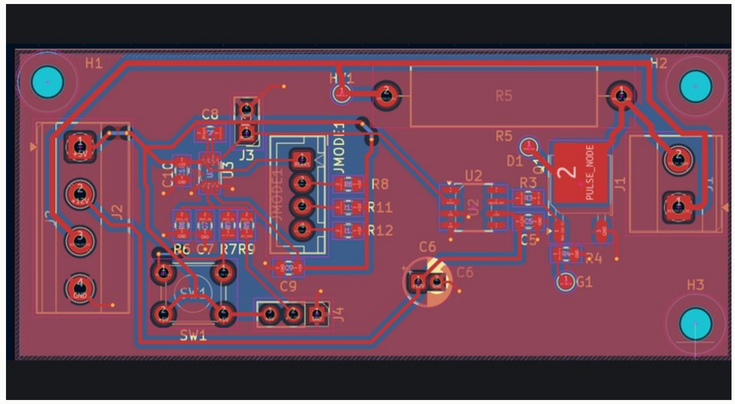
# Pulse Generator - Board + V1 Learning

May 2026 - Present / Research Hardware / PCB

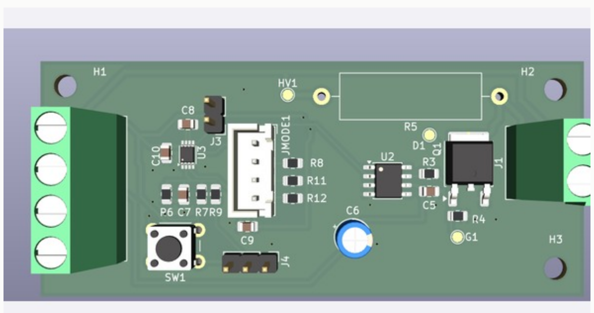
V1 microsecond hardware validated the basic bench workflow and exposed why the current board needs cleaner triggering, lab I/O, and grounding.



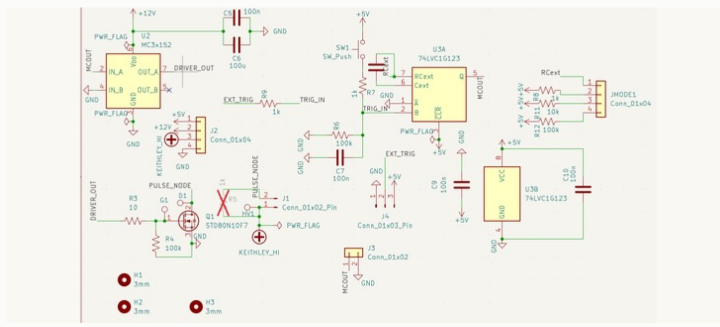
V1 microsecond board during bench probing. Useful as a learning artifact, not the current ns revision.




V1 PCB layout used to inspect trace routing, connector placement, and returns.




V1 rendered PCB for physical integration reference.



V1 microsecond schematic, included only as a predecessor to the current ns/us board.

**V1 EXPOSED**

- Clip-lead power and weak lab I/O made the first version awkward to use around real bench equipment.
- Triggering needed to be integrated with Python/RP2040 control instead of treated as an isolated pulse node.
- LTspice transient work was useful for V1 timing intuition, but the current PDF does not use the screenshot because it reads poorly.
- Grounding and connector strategy had to be designed as part of the experiment, not left to bench improvisation.

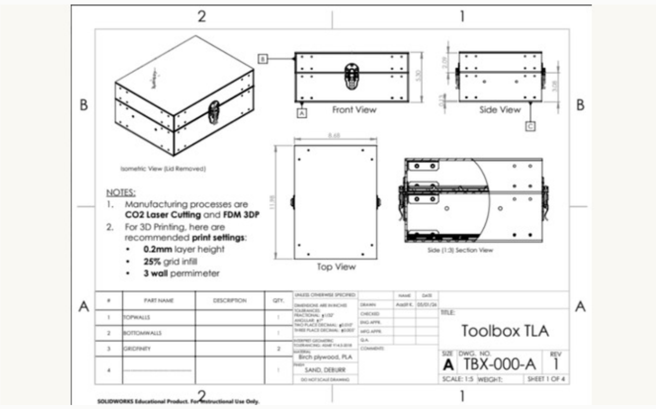
**CURRENT REVISION ADDS**

- Nanosecond and microsecond paths in one cleaner instrument.
- Coax/BNC-style outputs, shared sync/trigger handling, source-measure biasing, and clearer return paths.
- Validation plan: rails, trigger logic, dummy loads, switching behavior, then device-side fixture measurements.

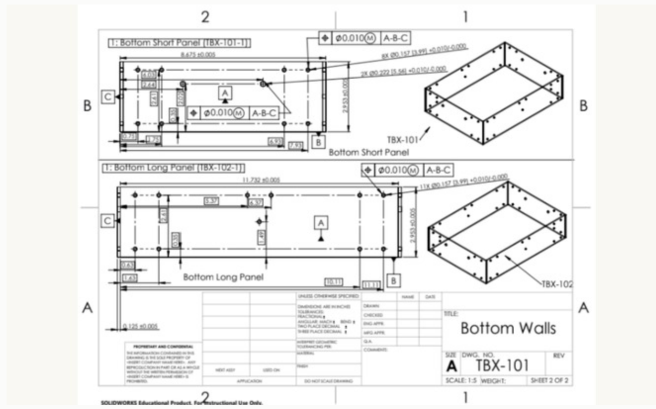
# Custom Toolbox Design + Manufacturing

May 2026 - Jun 2026 / Mechanical Design

Controlled CAD-to-drawings-to-fabrication exercise: model the assembly, define functional datums, create buildable drawings, fabricate parts, and inspect fit-up.




Assembly drawing defining lid, latch, hinge, body, and insert interfaces.




Part drawing with datum references and positional tolerance callouts.

Prototype fit-up after fabrication with hinges, latches, handle, and fasteners.

Open prototype showing interior fit-up; final-version photos unavailable.

### DESIGNED

- Production-style drawings using ASME Y14.5-2018 GD&T conventions.
- Datum scheme, position tolerances, fastener clearances, bend/edge reliefs, hinge/latch interfaces, and lid closure.
- Laser-cut plywood body with FDM/Gridfinity-style inserts and fastened joints visible in the prototype.

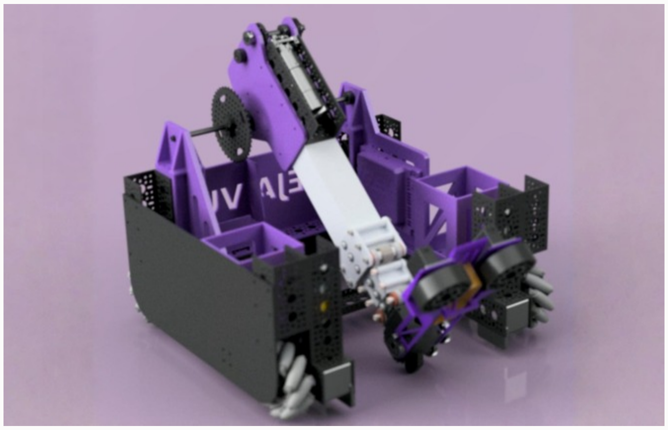
### INSPECTED

- Fit-up checked with calipers, fasteners/gauge pins, squareness checks, lid closure, and hinge/latch alignment.
- The value is drawing discipline and datum strategy, not pretending this was a metal production enclosure.
- Prototype photos are included honestly because final-version photos are not available.

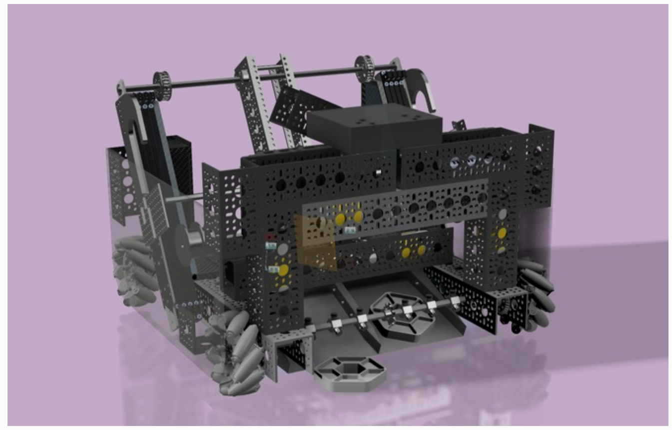
# FIRST Robotics Hardware

Aug 2021 - Jun 2025 / Robotics Hardware / Controls

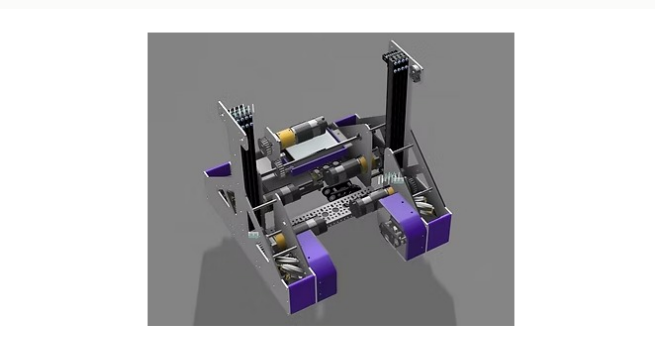
Four seasons of FTC hardware: 8 robots, 500+ part CAD assemblies, mechanism iteration under match constraints, Java controls, and three Robot Design Awards.



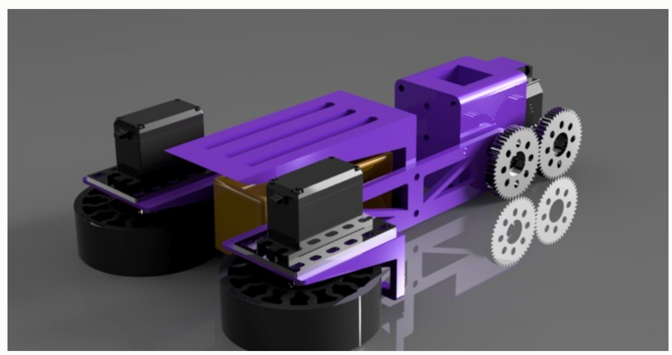
INTO THE DEEP robot CAD with drivetrain, active intake, transfer path, lift, and end effector.



CENTERSTAGE robot CAD assembly with pixel scoring mechanisms.



POWERPLAY robot CAD with lift and cone manipulation systems.



High-iteration intake subsystem CAD from Deja Vu INTO THE DEEP.

### BUILT ACROSS SEASONS

- Mecanum drivetrains, active intakes, transfer paths, arms/lifts, claws/end effectors, hang mechanisms, and game-specific scoring systems.
- FEA-informed custom aluminum and printed components for stress, deflection, and mechanism clearance.
- Manufacturing through printed parts, laser-cut plates, CNC-machined aluminum, and shop-fabricated hardware.

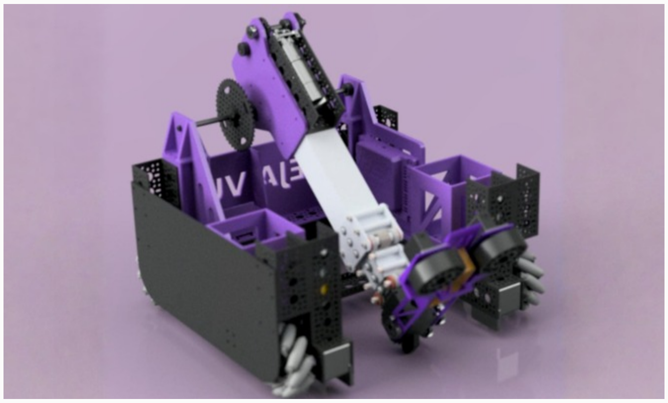
### CONTROLS-AWARE DESIGN

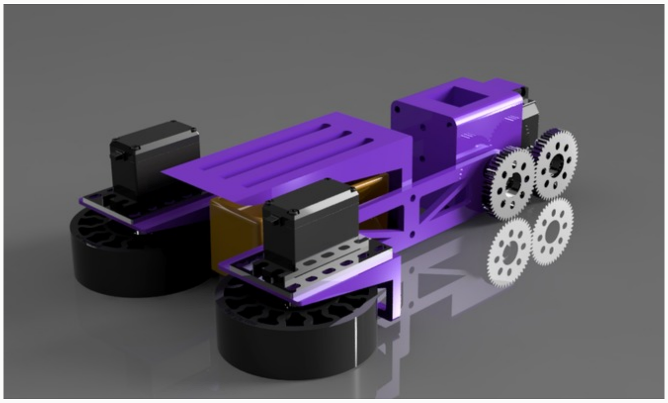
- Java TeleOp with field-centric mecanum, encoder PID loops, mechanism sequencing, driver feedback, and autonomous routines.
- Hardware and driver code were tuned together so mechanisms were not only possible in CAD but usable under match pressure.
- Mentored 30+ members in CAD and Java; team-building is secondary here, but it affected design execution.

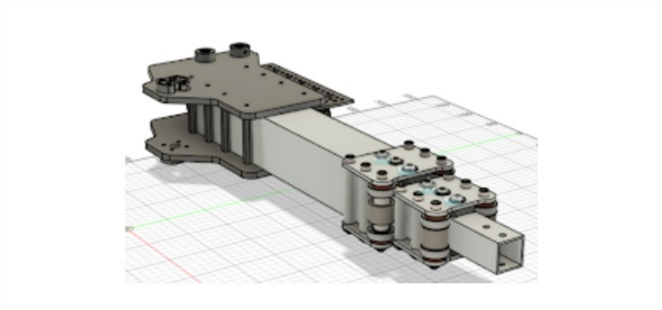
# Deja Vu - INTO THE DEEP

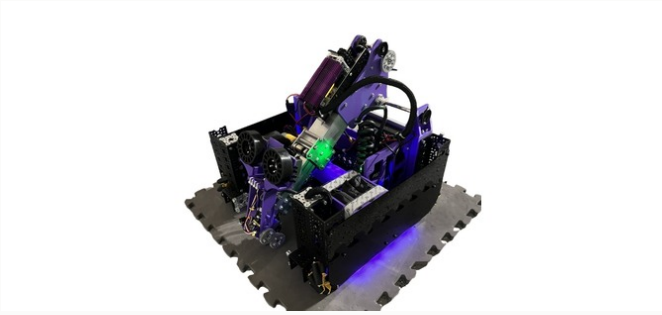
Aug 2024 - Jun 2025 / FTC Robotics / Hardware Lead

500+ part CAD robot for sample pickup, basket scoring, specimen scoring, and endgame hang, designed under real match reliability constraints.


Full robot CAD with intake, transfer, lift, end effector, drivetrain, and hang packaging.


High-iteration intake mechanism CAD.


End-effector/linkage packaging detail.


Physical robot reference for mechanism packaging and service access.

### DESIGNED

- Owned mechanical design, CAD management, and manufacturing coordination for FTC #13216.
- Four intake revisions: passive funnel, surgical-tubing roller, dual-stage roller, then a simpler high-speed roller geometry.
- Designed mecanum drivetrain, active intake, transfer pathway, arm/lift, end effector, and hang hardware.

### TECHNICAL WORK

- FEA on arm pivot and hang brackets; checked stress, safety factor, and deflection against mechanism travel.
- Manufactured with Markforged nylon, Prusa prototypes, laser-cut polycarbonate/Delrin, CNC aluminum, and shop tools.
- Wrote Java TeleOp with field-centric drive, arm/lift PID, intake sequencing, and driver rumble feedback.

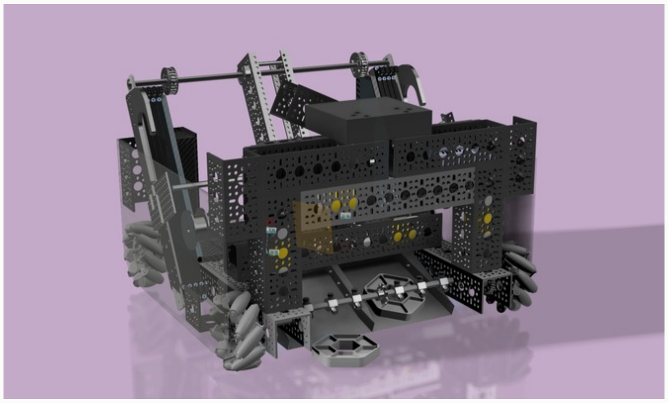
### TOOLS

SolidWorks / FEA / Java / FTC SDK / PID Control / Mecanum Kinematics / Laser Cutting / Markforged

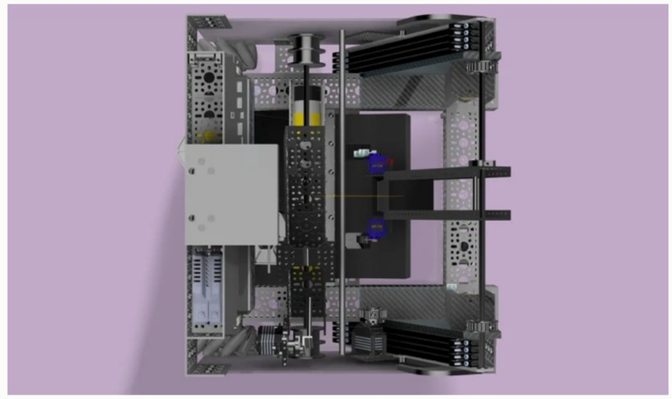
# Deja Vu - CENTERSTAGE

Aug 2023 - Jun 2024 / FTC Robotics / Hardware Lead

Pixel-scoring robot for FTC CENTERSTAGE, focused on reliable dual-pixel handling, backdrop scoring, vision alignment, and driver-controlled mechanism sequencing.



Full robot CAD assembly for CENTERSTAGE.



Robot/app or subsystem reference from the season.

#### DESIGNED

- Compliant TPU claw fingers, passive spring-loaded grip, and servo-actuated release for hexagonal pixels.
- Dual-pixel capacity to cut cycles to the backdrop; geometry had to grip without blocking the vision workflow.
- Robot packaging around mecanum drive, arm presets, claw actuation, and paper-airplane endgame launcher.

#### CONTROLS + RESULTS

- AprilTag-based autonomous alignment and camera-based pixel-position feedback.
- Java TeleOp with speed curves, arm presets, claw control, drone launcher arming, and intake/transfer coordination.
- Qualified for NorCal regional championship with minimal mechanical failures during matches.

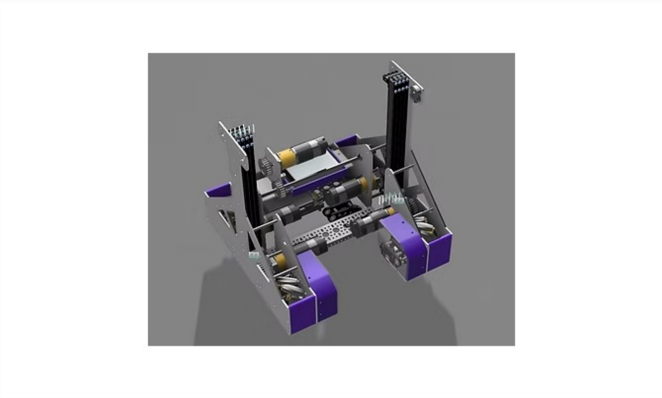
#### TOOLS

SolidWorks / Java / FTC SDK / AprilTags / OpenCV / PID Control / Mecanum Kinematics

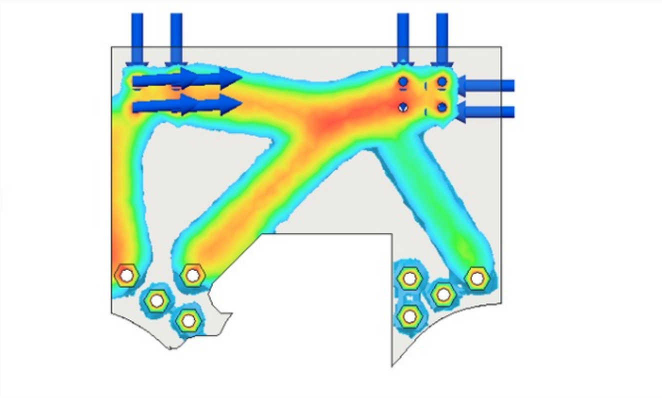
# Zenith - POWERPLAY

Aug 2022 - Jun 2023 / FTC Robotics / Team Lead

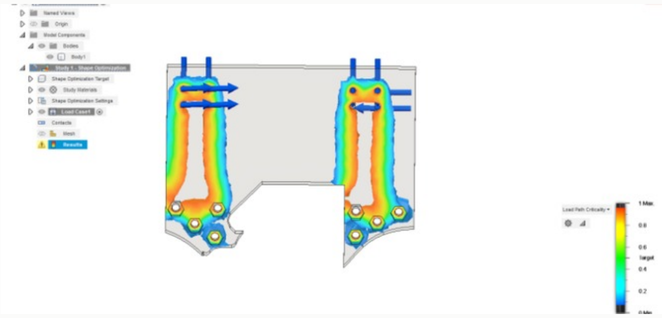
Cone-stacking robot built around a fast linear lift, cone-centering claw, mecanum drive, and autonomous positioning.



POWERPLAY robot CAD with lift and cone-scoring mechanism.



Mechanism/simulation reference for the season.



Additional CAD/simulation view.

Robot or subsystem image from Zenith.

### BUILT

- Linear lift with 30+ inches of travel, sub-1.5s extension target, and about 200g carried load.
- String-rigged lift using dyneema and spool geometry balanced against motor torque and cycle time.
- Three claw revisions, landing on internal centering guides before gripping.

### CONTROLS + LESSONS

- Java TeleOp with mecanum kinematics, lift PID, and speed scaling modes.
- Autonomous used OpenCV signal-sleeve detection, dead-wheel odometry, and IMU heading correction.
- ZenLender spreadsheet lending system started here and later evolved into Inventry.

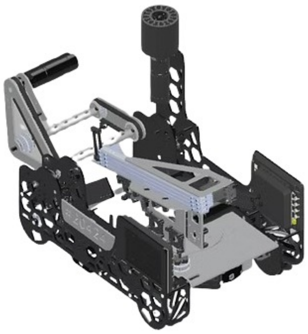
### TOOLS

Onshape / Java / FTC SDK / OpenCV / Dead-Wheel Odometry / PID Control / String Rigging

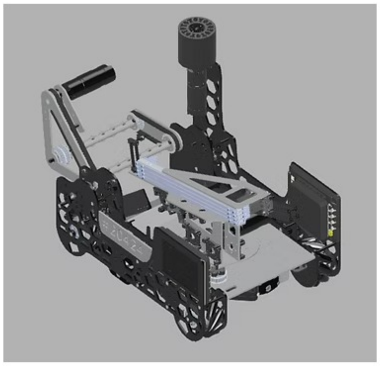
# Zenith - FREIGHT FRENZY

Aug 2021 - Jun 2022 / FTC Robotics / First Season

First FTC season on Zenith #20424: learned CAD, manufacturing, basic Java control, and robot subsystem assembly through a freight-scoring robot.



Freight Frenzy robot image.



Freight Frenzy robot CAD/build reference.

#### LEARNED

- Onshape constraints, simple brackets, subassemblies, and complete robot packaging basics.
- Drill press, bandsaw, 3D printing, hole placement, print warping, and basic shop workflow.
- Motor-controller wiring and basic mechanism integration through a functioning FTC robot.

#### SOFTWARE

- Java gamepad input handling, arm/intake motor control, encoder-based autonomous moves, and carousel timing.
- Code was simple and hard-coded, but it established the foundation for later PID/state-machine work.
- Robot scored freight, delivered ducks, and parked successfully in matches.

#### TOOLS

Onshape / Java / FTC SDK / Encoder Autonomous / 3D Printing / Drill Press / Bandsaw

# Serenity - FREIGHT FRENZY

Aug 2021 - Jun 2022 / FTC Robotics / Team Lead

Led a newer FTC team while also learning on Zenith, building team process and a reliable Freight Frenzy robot.


Serenity robot/team visual asset from the website.

**BUILT**

- Tank-drive robot with basic arm for freight placement, passive intake with compliant wheels, and carousel spinner.
- Design priority was reliability and match completion rather than maximum mechanism complexity.
- Competed in league meets with a functioning robot for a young team.

**TEAM PROCESS**

- Started weekly design reviews, shared CAD workspace conventions, manufacturing priority lists, and notebook assignments.
- Mentoring while still learning forced me to explain CAD, build, and competition decisions clearly.
- This was the first time I built team infrastructure, not only robot parts.

**TOOLS**

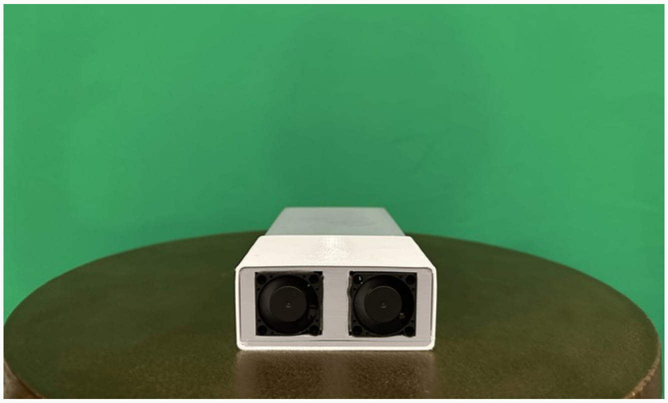
Onshape / Robot CAD / Manufacturing Planning / Engineering Notebook / Tank Drive / Carousel Mechanism

# Construction Acclimation

Oct 2022 - Apr 2023 / Hardware / Data Logging

$47 concrete-curing temperature/humidity probe and dashboard prototype intended as a low-cost alternative to commercial construction acclimation devices.

Probe prototype hardware.



Electronics/enclosure or dashboard context.

**BUILT**

- Arduino Nano, DHT22, DS18B20, RTC, MicroSD logging, LiPo battery, fans, and PETG enclosure.
- Logged readings every 5 minutes with redundant temperature measurement.
- Client-side dashboard for CSV upload, trend plotting, out-of-range flags, and PDF reporting.

**TESTED**

- Compared against a calibrated Humboldt H-4210 commercial unit for 14 days.
- Temperature deviation about +/-0.8C, humidity about +/-3.2%, and 4,028/4,032 readings captured.
- Presented to City of Palo Alto officials; certification/liability/durability remained open issues.

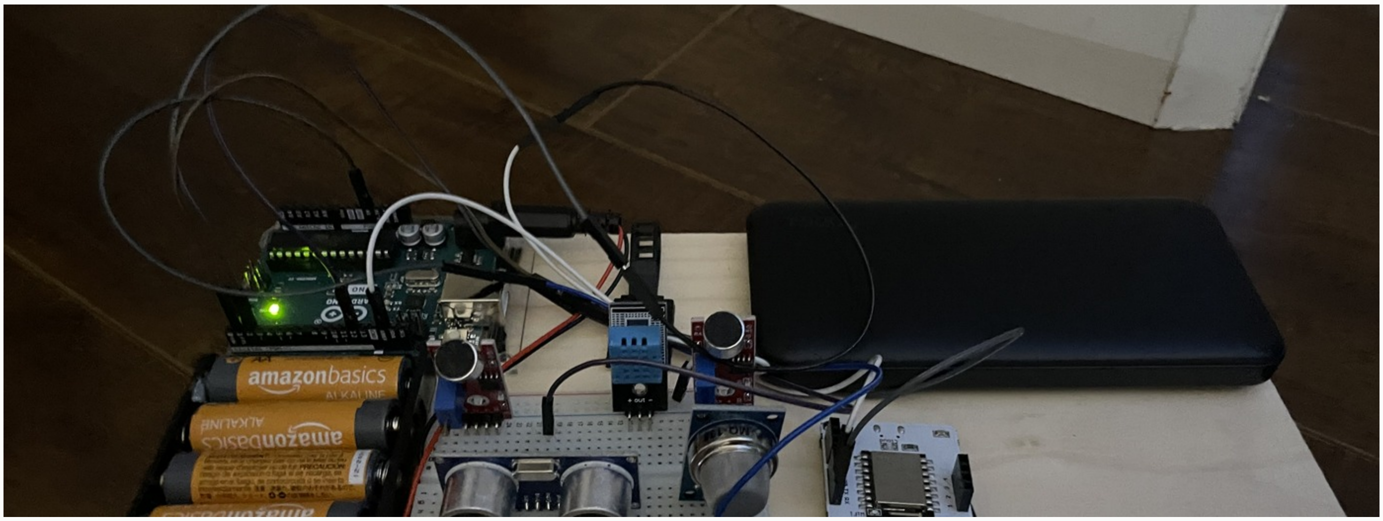
**TOOLS**

Arduino Nano / DHT22 / DS18B20 / RTC / MicroSD / Chart.js / PETG Enclosure

# Friendly Fires

Sep 2021 - Mar 2022 / Sensor Hardware

Coal-seam fire detection prototype using temperature, gas, and alert sensors integrated into a low-cost safety device concept.




Functional prototype electronics with sensors, power, microcontroller, and breadboard wiring.

### BUILT

- Arduino-based sensor array for temperature, carbon monoxide, methane, and oxygen-depletion monitoring.
- Haptic, LED, and optional audio alerts intended for low-light mining environments.
- Rechargeable battery and low-power sleep-mode concept for a 12+ hour work shift.

### PROTOTYPE STATUS

- Demonstrated controlled CO detection, temperature-gradient detection, and multi-sensor integration.
- Proof of concept only: no mining-environment calibration, false-positive study, dust/humidity durability, or regulatory approval.
- Included as an early sensor-integration project, not a deployable safety product.

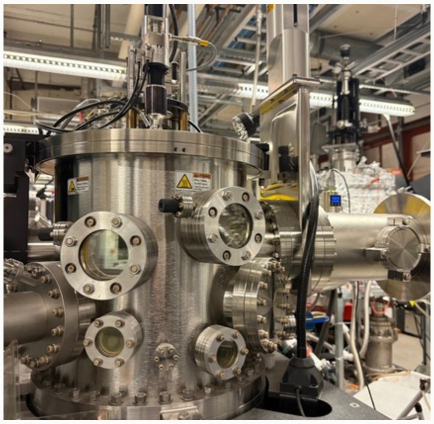
### TOOLS

Arduino / CO Sensor / Methane Sensor / IR Temperature / Haptic Alerts / Sensor Integration

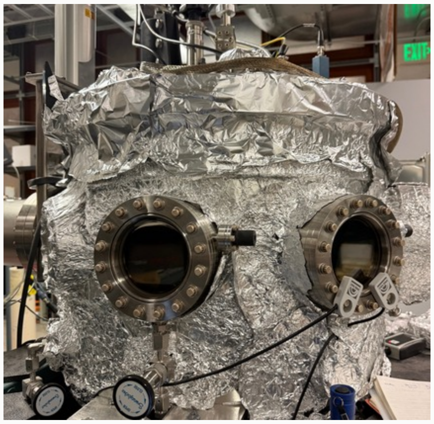
# Ramesh Lab: Complex-Oxide Thin Films

Jan 2026 - Present / Research

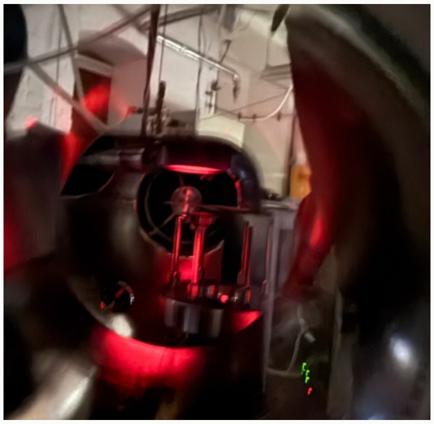
Researching $BiFeO_3$ and related complex-oxide thin films for faster, lower-power memory and logic beyond DRAM and CMOS.



Lab/growth workflow context.



Thin-film process and sample handling context.



Experimental hardware context from lab work.

### LEARNING + GROWTH WORK

- Reading Lines and Glass to build the physics base: polarization, domains, coercive fields, hysteresis, dielectric response, strain, and defects.
- Learning supervised substrate prep, solvent cleaning, annealing, step-terrace control, PLD growth, and RHEED monitoring.
- Practicing thin-film deposition workflow with Donald's guidance: oxygen pressure, plume shape, target/substrate geometry, and temperature stability.
- Connecting growth choices to AFM, XRD, PFM, and electrical characterization results.

### CONNECTED PROJECTS

- Nanosecond Pulse Generator: lab instrument for fast electrical characterization of oxide devices after fabrication.
- PLDTracker: data-management tool linking deposition conditions to characterization data and lab documentation.
- The point is the full research loop: grow films, track process parameters, then measure how the devices switch.

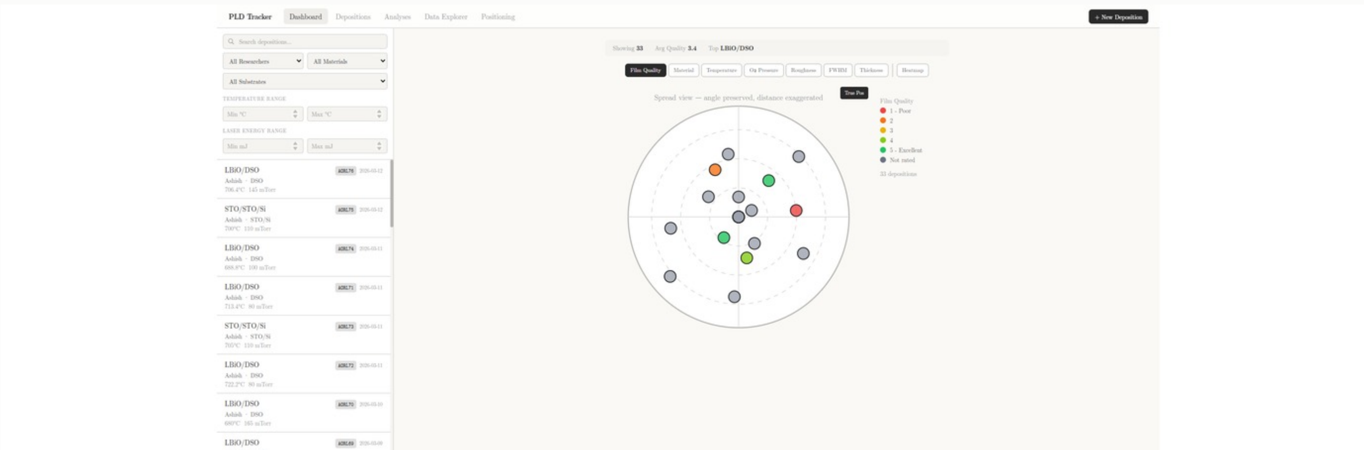
### WHY IT MATTERS

Strain, interfaces, atomic stacking order, and oxygen vacancies can create switchable polarization, interfacial conduction, and coupled ferroic states that do not exist in a single bulk material. The work is about learning how process choices reshape device behavior.

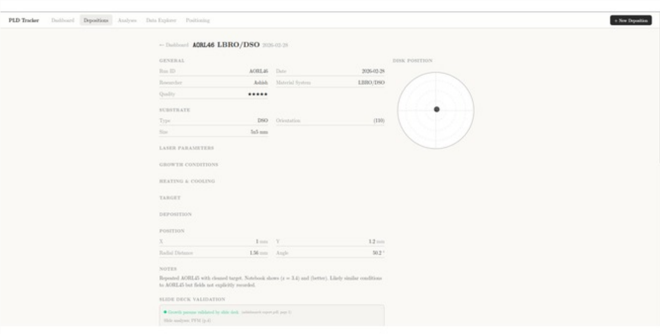
# PLDTracker (Ramesh Lab)

Jan 2026 - Mar 2026 / Research Software

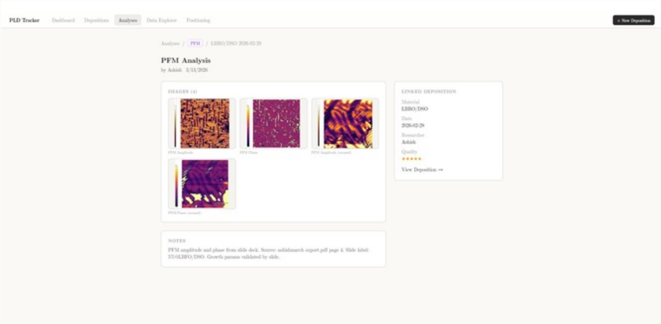
Experiment tracking and data-management system for PLD thin-film workflows, built to connect growth parameters, wafer position, characterization images, and lab documentation.




Dashboard and deposition data explorer for tracking thin-film growth records.



Filtering and data exploration interface.



Analysis-image/document workflow tied back to deposition records.

### IMPLEMENTED

- Logs substrate, target material, temperature, oxygen pressure, laser fluence, deposition time, and wafer disk position.
- Links XRD scans, AFM/PFM domain images, slide-deck figures, and camera captures to specific runs.
- Built a dashboard for filtering records and visualizing parameter trends with connected datapoints and fit lines.

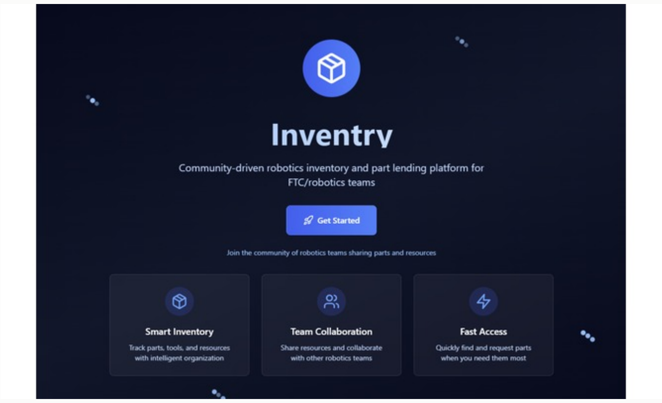
### RESEARCH VALUE

- Replaces scattered notes, spreadsheets, screenshots, and slide decks with searchable experiment metadata.
- Makes it easier to revisit whether growth-condition changes explain film-structure or device-property changes.
- Includes a SolidWorks camera mount path for substrate-position monitoring at the PLD chamber.

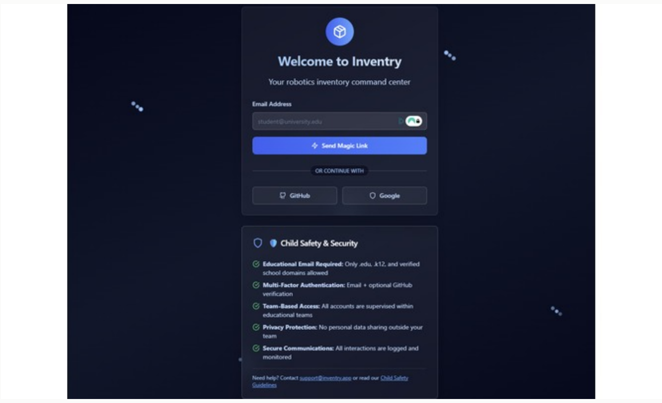
# Inventry

Jun 2025 - Dec 2025 / Robotics Software

Robotics-parts inventory, lending, and marketplace system built to reduce emergency shipping and scattered spreadsheet workflows for FTC teams.




Inventry landing/interface screenshot.




Inventory/product interface screenshot.

**IMPLEMENTED**

- Inventory management for parts, quantities, condition, team ownership, and search.
- Part lending network with distance matching, urgent requests, and lender reputation.
- Marketplace path for robotics components, with team verification through FIRST data.

**TECHNICAL WORK**

- React/TypeScript frontend, Node/Express backend, PostgreSQL data model, Vercel/Railway hosting.
- Invoice parsing with Tesseract OCR and GPT-4 structured extraction for part numbers, quantities, and prices.
- Weekly scrapers for GoBilda, REV, ServoCity, and AndyMark with 3,000+ SKU search data.

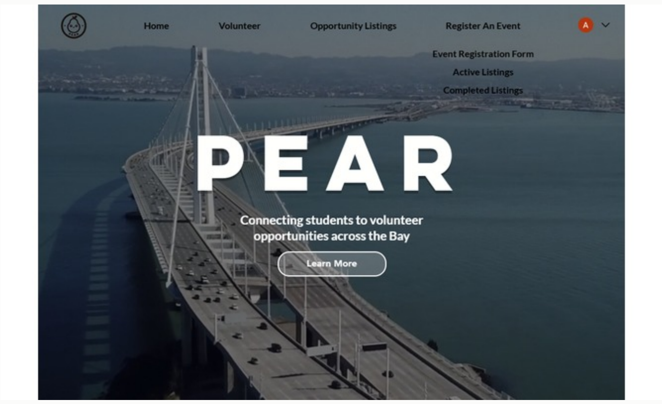
**TOOLS**

React / TypeScript / Node.js / Express / PostgreSQL / Tesseract OCR / Web Scraping / OAuth 2.0

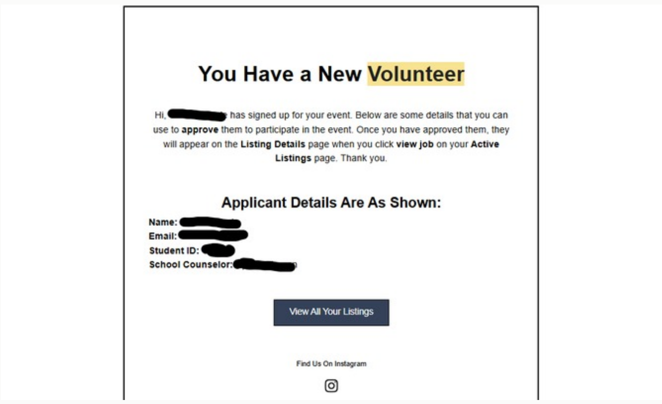
# Pear Volunteering

Aug 2023 - May 2025 / Web Platform

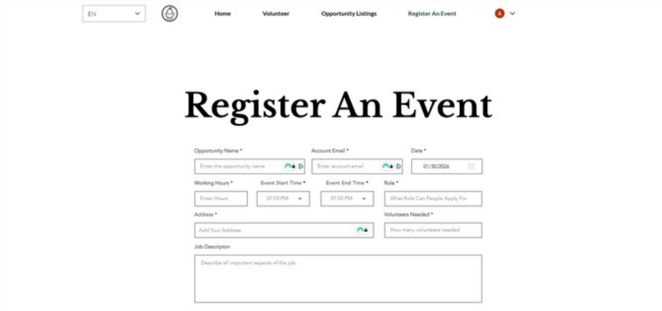
Volunteer platform connecting 2000+ students, 20+ organizers, and administrators who needed verified hour logs.



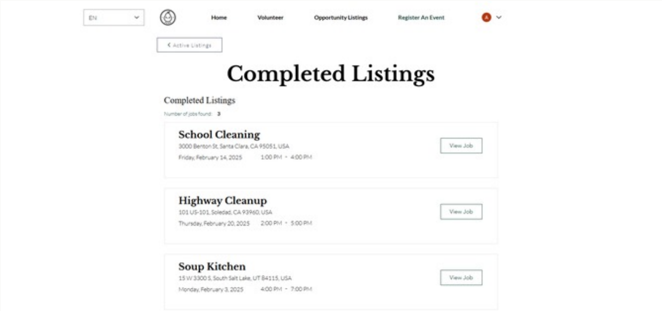

Pear volunteer/event platform screenshot.




Volunteer interface or organizer screen.




Supporting platform screenshot.




Volunteer workflow screenshot.

**IMPLEMENTED**

- Student event browsing, organizer event creation, administrator verification, and hour-log workflows.
- Student ID validation, schedule-conflict checks, capacity updates, organizer notifications, and QR check-in.
- Auto-generated verification letters as PDFs for documented service hours.

**TECHNICAL WORK**

- Wix Velo backend logic, custom JavaScript, webhook integrations, and role-based access.
- Async webhook queue for high-volume signup events that could fill 50+ spots quickly.
- Privacy constraints: student contact data stayed protected and communication flowed through the platform.

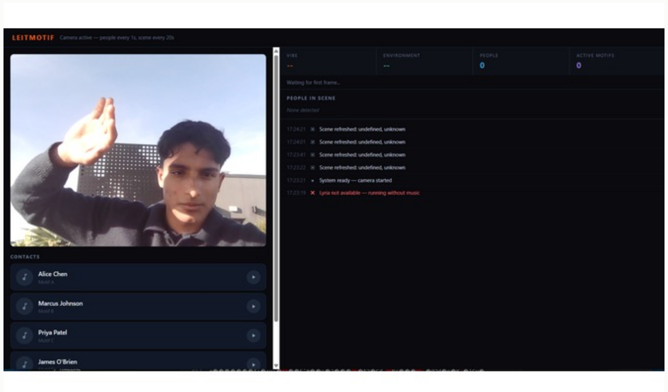
**TOOLS**

Wix Velo / JavaScript / Webhook Queues / QR Check-In / PDF Generation / Role-Based Access / Data Privacy

# Leitmotif

Mar 2026 / Hackathon / Accessibility Software

YC x DeepMind hackathon winner: real-time generative music interface for visually impaired users, assigning musical motifs to people in a scene.




Leitmotif web/app interface screenshot.

Hackathon or demo context.

#### IMPLEMENTED

- Camera loop sends frames to Gemini vision, extracts scene/person state, then updates the music engine.
- Each person gets a persistent musical motif with entry and exit cues.
- Crossfades emotional state changes over 4-6 seconds so scene changes do not sound abrupt.

#### ARCHITECTURE

- Two-server split: camera analysis and real-time music orchestration decoupled for different latency profiles.
- Gemini 2.5 Flash for vision/orchestration, Lyria RealTime for PCM audio generation, Supabase for persistence.
- React Native companion app with monitor, contacts, visualizer, and settings views.

#### TOOLS

TypeScript / Node.js / React Native / Gemini API / Lyria Realtime / Supabase / Computer Vision

# Tools + Methods

Current / Engineering Toolkit

A compact map of the tools and methods that show up across the project pages.

**CAD / MECHANICAL**

- SolidWorks
- Onshape
- Fusion 360
- GD&T
- DFM/DFA
- FEA
- mechanism packaging
- tolerance stackups

**MANUFACTURING**

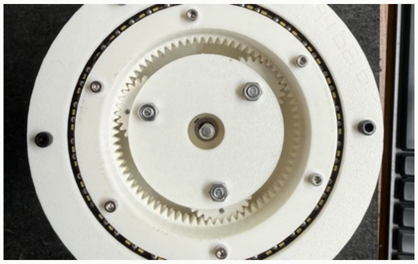
- mill
- lathe
- waterjet
- laser cutter
- TIG
- FDM/SLA
- Markforged
- fastener fit-up

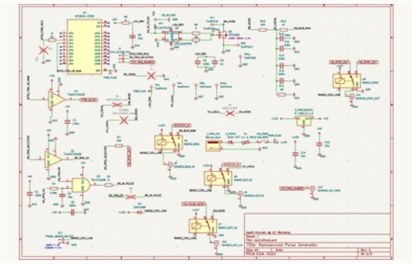
**ELECTRONICS / TEST**

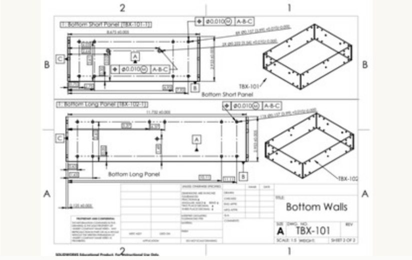
- KiCad
- LTspice
- oscilloscope
- source-measure
- RP2040/Python
- coax/BNC interfaces
- microcontrollers

**SOFTWARE / DATA**

- Python
- MATLAB/Simulink
- Java
- React/TypeScript
- Node.js
- PostgreSQL
- JupyterLab
- LabVIEW


Actuator prototype


PCB schematic



GD&T drawing